

Class 5

Vector Data Analysis

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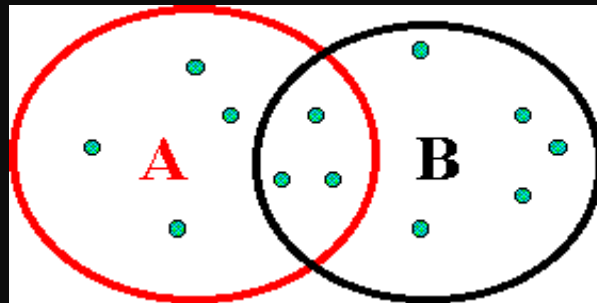
September 15th, 2026

Data issues in vector processing

- World is complex but data represents a simplified version
- In vector data, objects are represented as points, lines and polygons
- To get data into a GIS, two procedures are used,

Selection and Representation

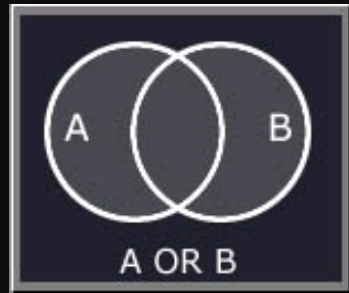
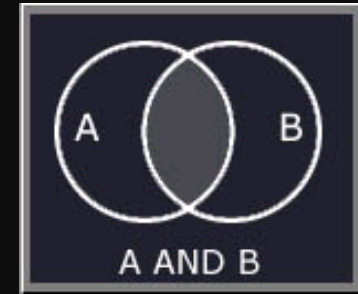
Overlay operations in a GIS



Computational Geometry

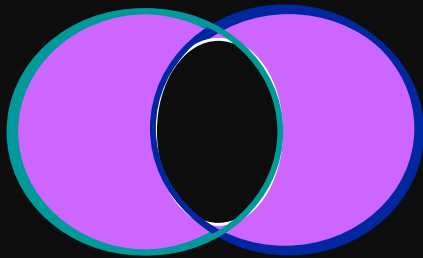
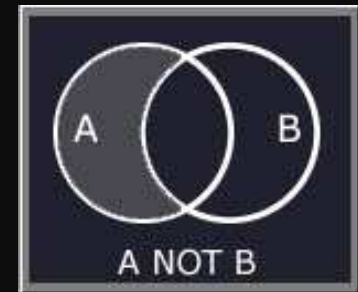
Boolean Logic in GIS

AND operator ($\cap$) is the intersection of two sets - those entities that belong to both set A and B ($A \cap B$)

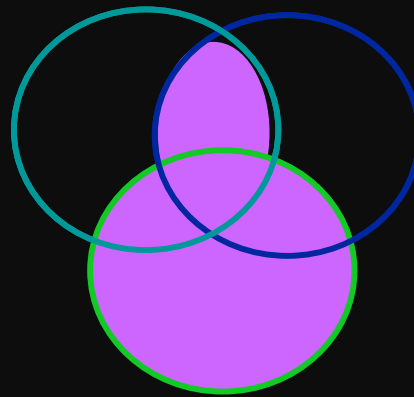


OR operator ($\cup$) is the union of two sets - those entities that belong to either set A or to set B ($A \cup B$)

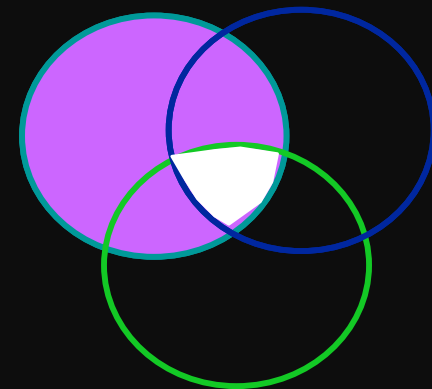
NOT operator ($\neg$) is the difference operator - identifying those entities that belong to A but not to B ($A \neg B$)



A XOR B



(A AND B) OR C



A AND (B OR C)

Classification of Vector Overlay Operations

Topological vector overlay operations - two methods:

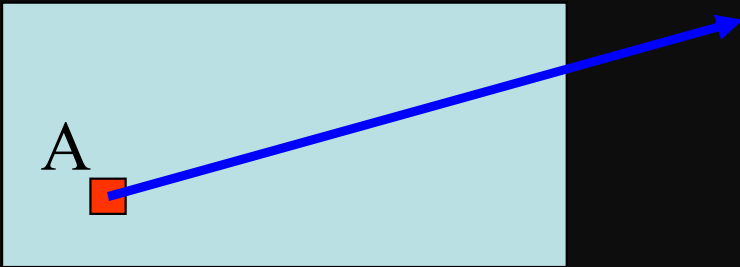
1. Through the **elements** contained in the layers to be overlaid
2. By **operation type**

For Element based method

Element types	Point	Lines	Polygons
Points	Points Coincide	Point in Line	Point in Polygon
Lines	Point in Line	Line intersection	Line in Polygon
Polygons	Point in Polygon	Line in Polygon	Polygon Overlay

Simplest form of overlay

Point in polygon procedure



Line intersects
1 edge of polygon

Odd number of
Intersections = inside,
Even = outside

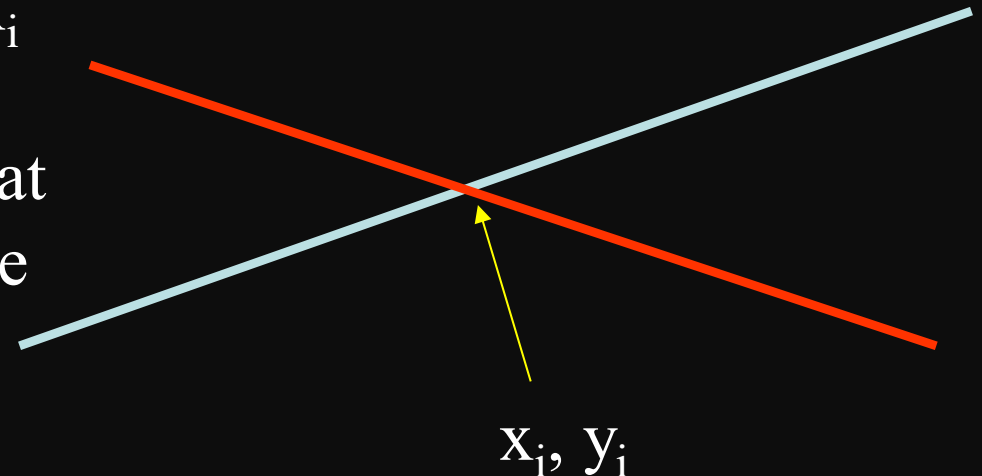
Count how many
intersections of the ray,
originating at point A,
pass through edges of
the polygon

$y = a1 + b1x$ and $y = a2 + b2x$
intersect at:

$$x_i = - (a1 - a2) / (b1 - b2), y_i = a1 + b1x_i$$

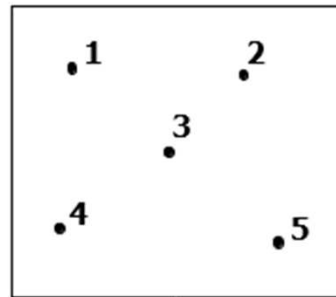
And check for the values of x_i to see that
it falls within the x values of each of the
lines.

Line intersections



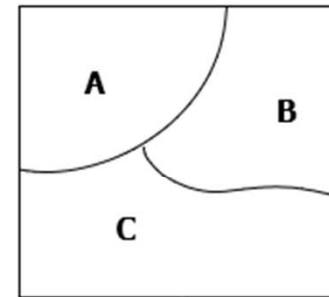
Point in polygon

Restaurant

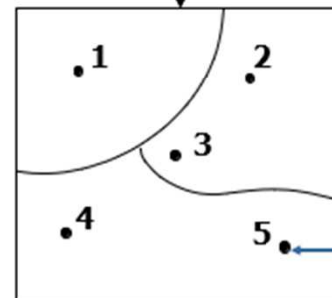


ID	Restaurant
1	McDonalds
2	Pizza Hut
3	KFC
4	<u>Nirulas</u>
5	Café Coffee Day

Towns

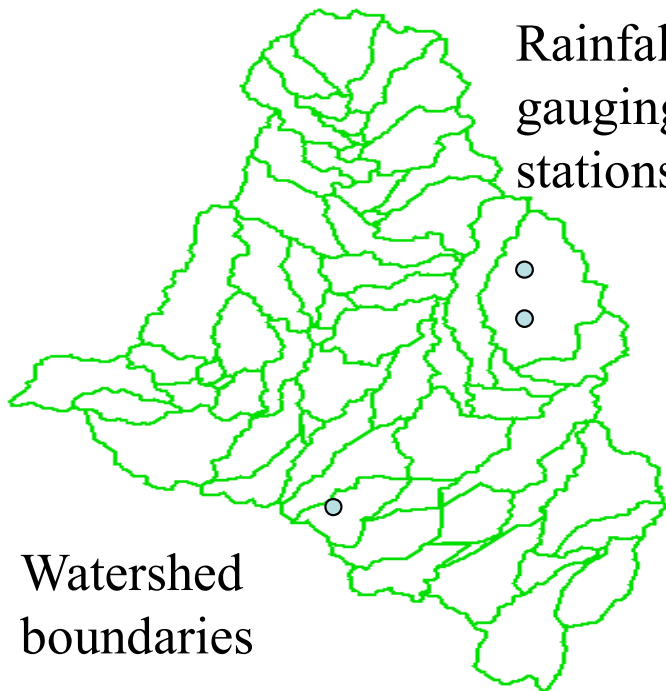


ID	City Ward
A	Gachibowli
B	Telecom Nagar
C	Anjaiah Nagar



ID	City Ward	Restaurant
1	Gachibowli	McDonalds
2	Telecom Nagar	Pizza Hut
3	Telecom Nagar	KFC
4	Anjaiah Nagar	<u>Nirulas</u>
5	Anjaiah Nagar	Café Coffee Day

Rainfall
gauging
stations

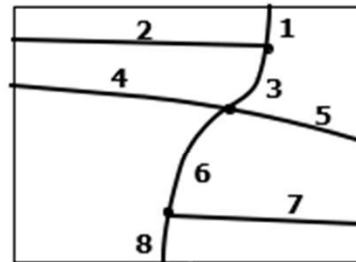


Watershed
boundaries

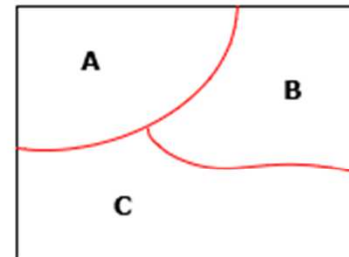
Line in polygon

ID	Highway Number
1	NH03
2	NH12
3	NH03
4	NH07
5	NH07
6	NH03
7	NH25
8	NH03

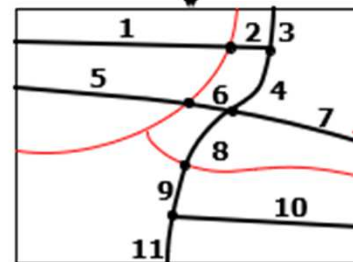
Roads



Soil Type



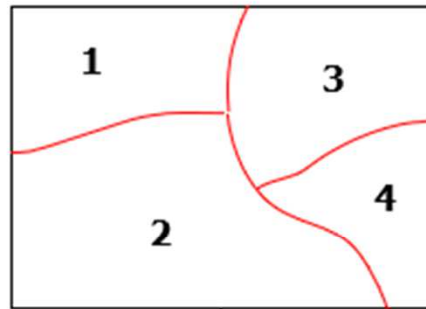
ID	Soil Type
A	<u>Moorum/Redl</u>
B	Laterite
C	Alluvial



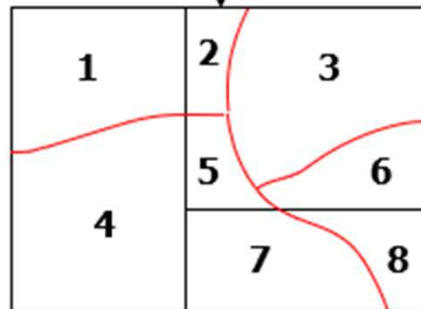
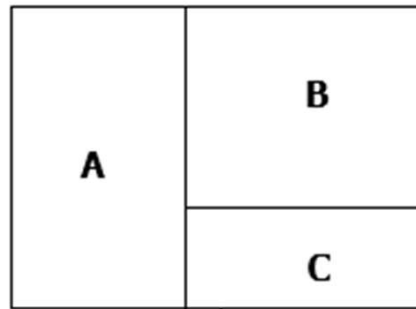
ID	Original Line ID	Highway Number	Soil Type
1	2	NH12	<u>Moorum/Red</u>
2	2	NH12	Laterite
3	1	NH03	Laterite
4	3	NH03	Laterite
5	4	NH07	<u>Moorum/Red</u>
6	4	NH07	Laterite
7	5	NH07	Laterite
8	6	NH03	Laterite
9	6	NH03	Alluvial
10	7	NH25	Alluvial
11	8	NH03	Alluvial

Polygon overlay

Watershed

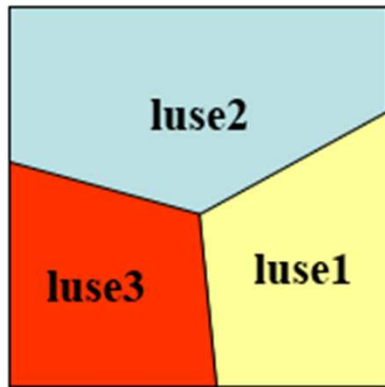


District

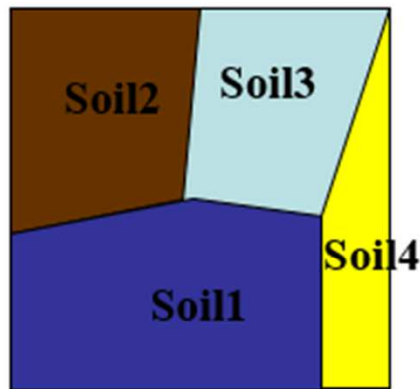
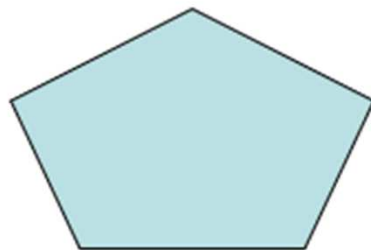


ID	Watershed	District
1	1	A
2	1	B
3	3	B
4	2	A
5	2	B
6	4	B
7	2	C
8	4	C

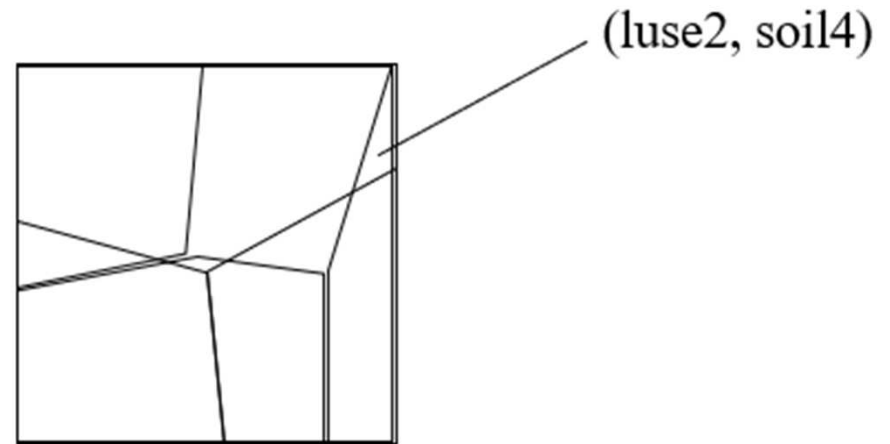
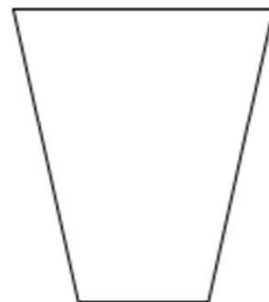
Polygon overlay



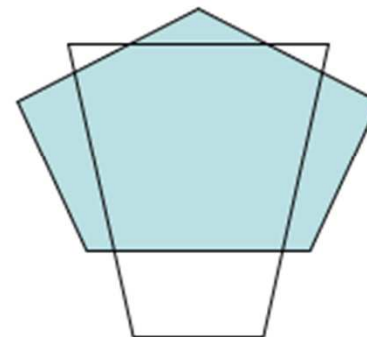
Landuse



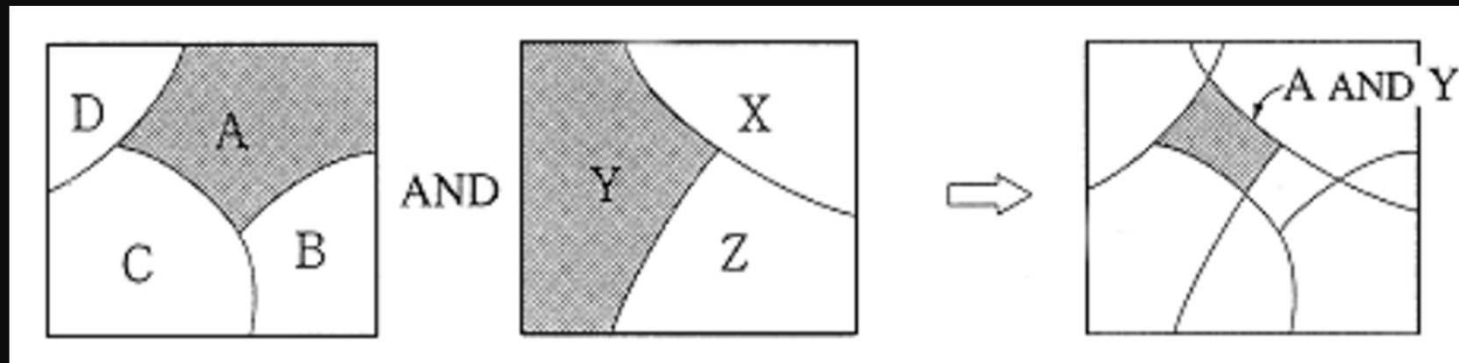
Soil



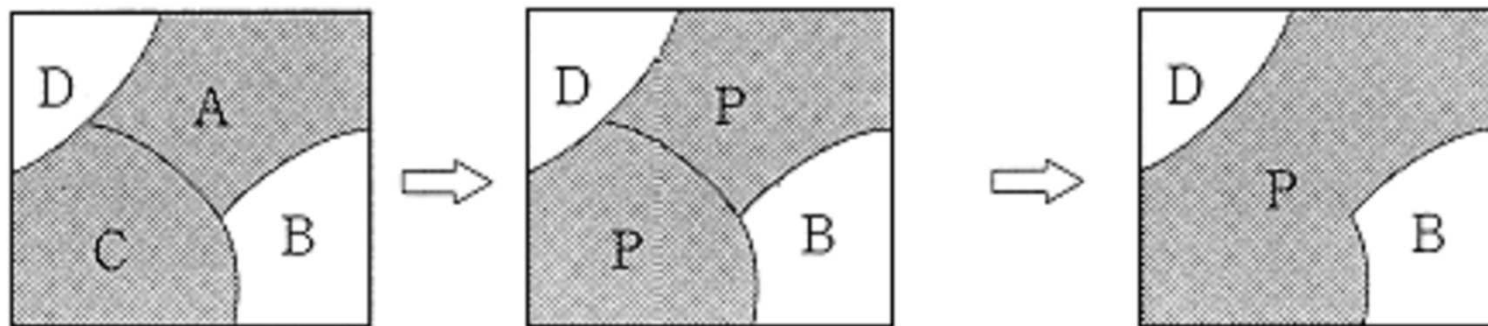
Field case: all polygons with
landuse and soil attributes



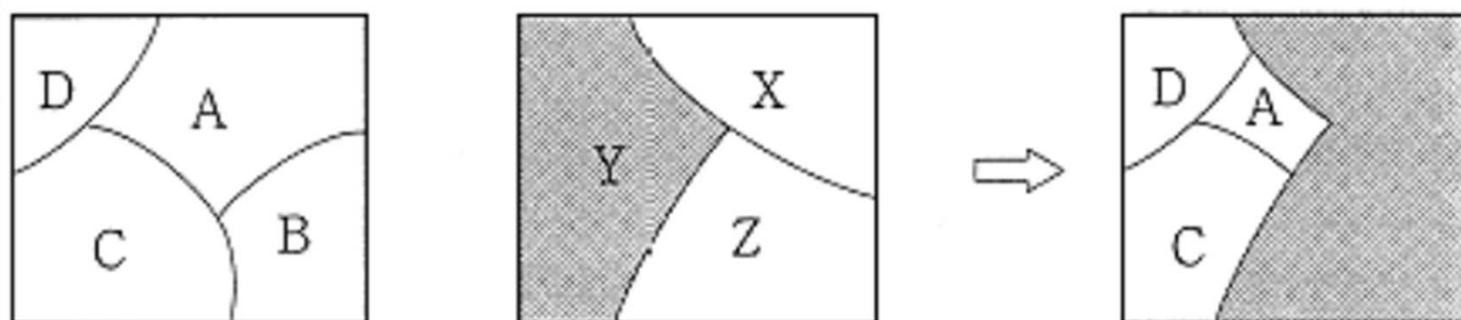
Discrete object case: 7 polygons after overlay



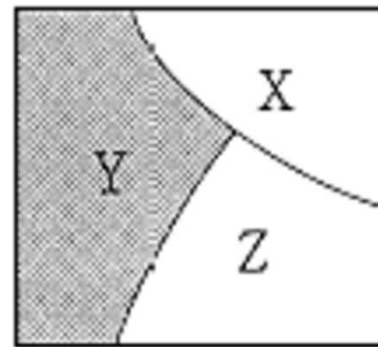
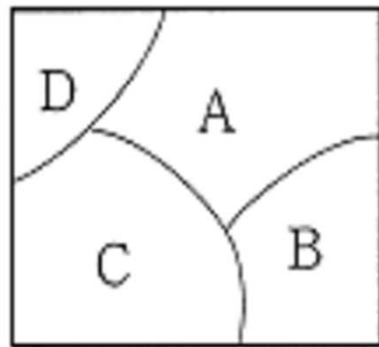
Query:
what is the area
of both A and Y?



Reclassify:
Reclassify A
and C into P

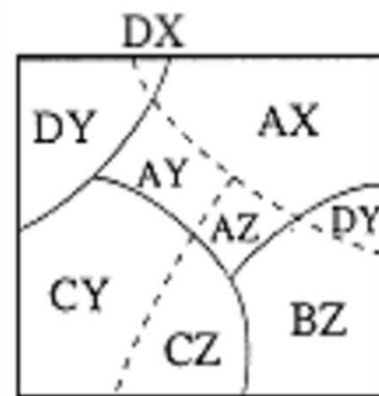


**Coverage
rebuilding:**
Clip onto Y zone
and erase other



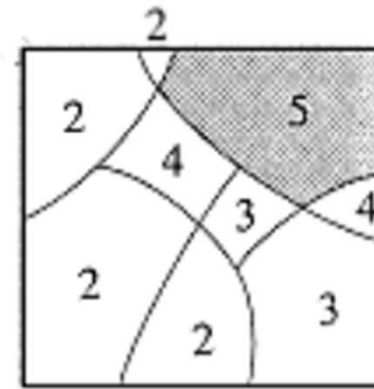
Overlay:

What are the ranks of overlaid areas?

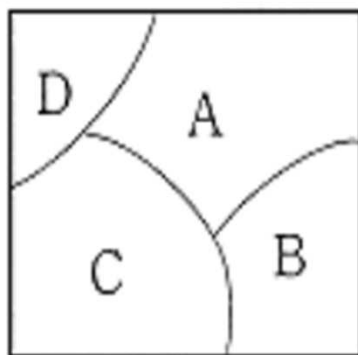


Ranking

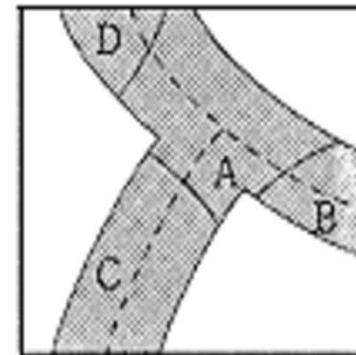
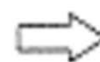
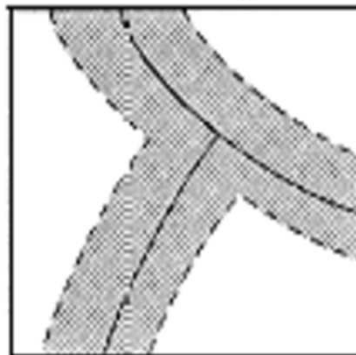
	A	B	C	D
X	5	4	3	2
Y	4	4	2	2
Z	3	3	2	1



Buffer zone around
road n/w



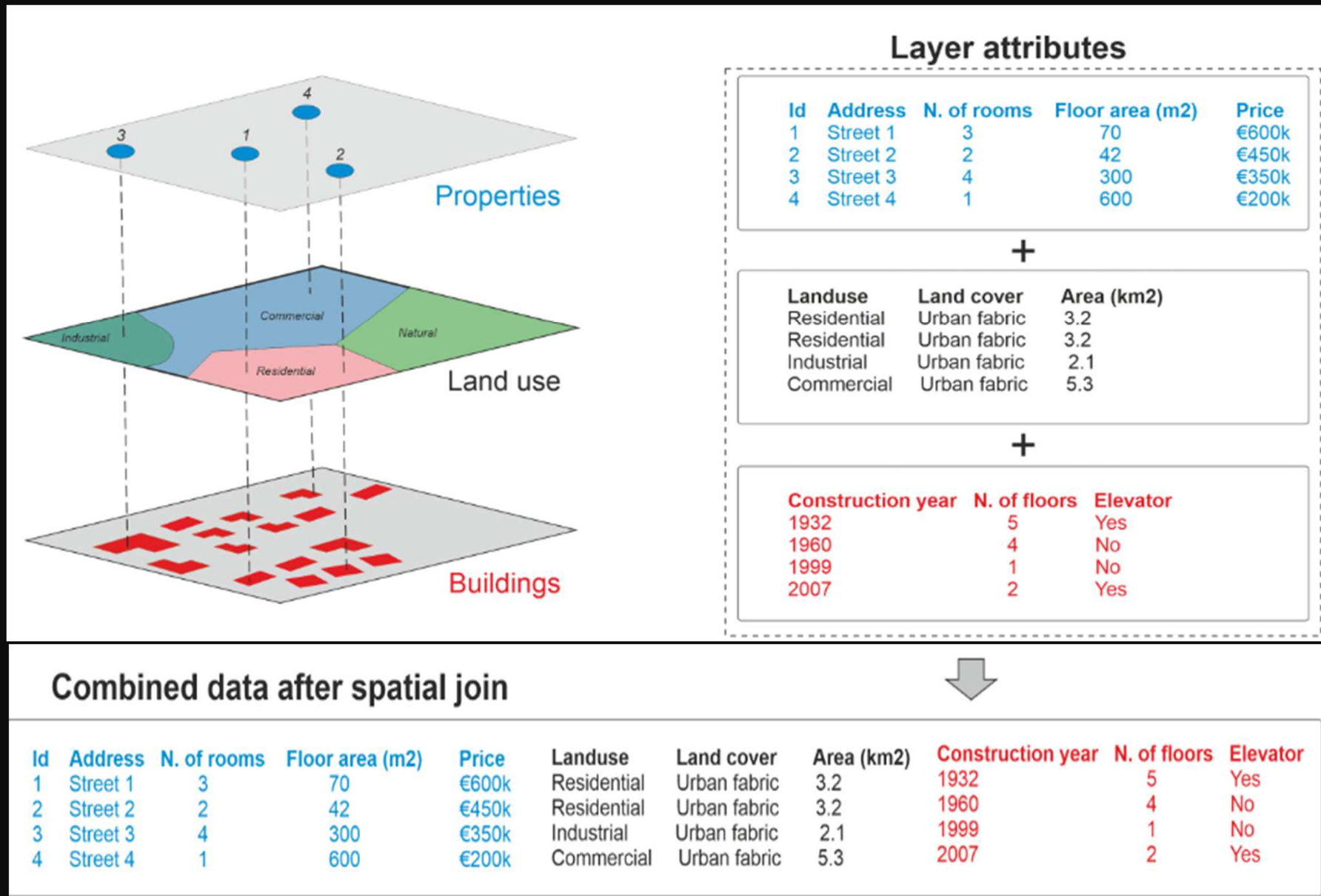
+



Connectivity:
Rezonify buffer
zone using
connectedness
with the overlaid
area

General overlay types

Spatial Join

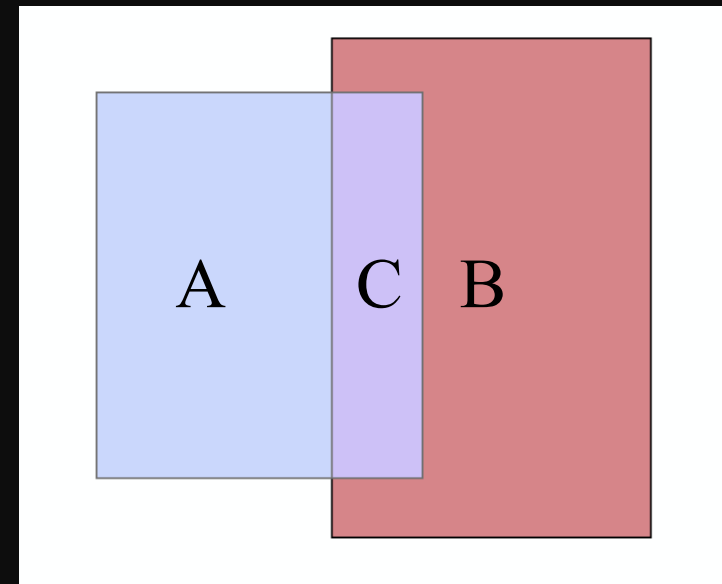
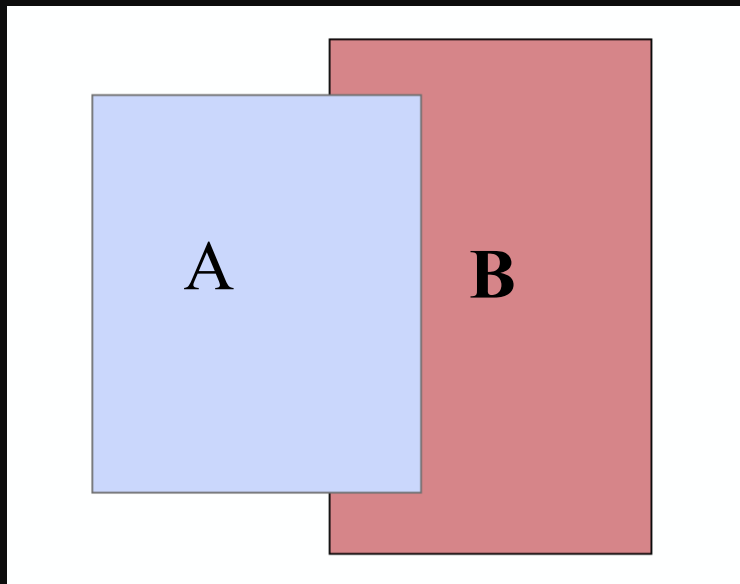


Spatial join allows you to combine attribute information from multiple layers based on spatial relationship (<https://pythongis.org/part2/chapter-06/nb/06-spatial-join.html>)

Clip

Two polygons, A and B, Overlap.
Clip A using B as a cookie cutter.

Clip operation creates a **new polygon**, C,
which is the intersect, or overlap, of A and B.
Attributes of B do not appear in C.

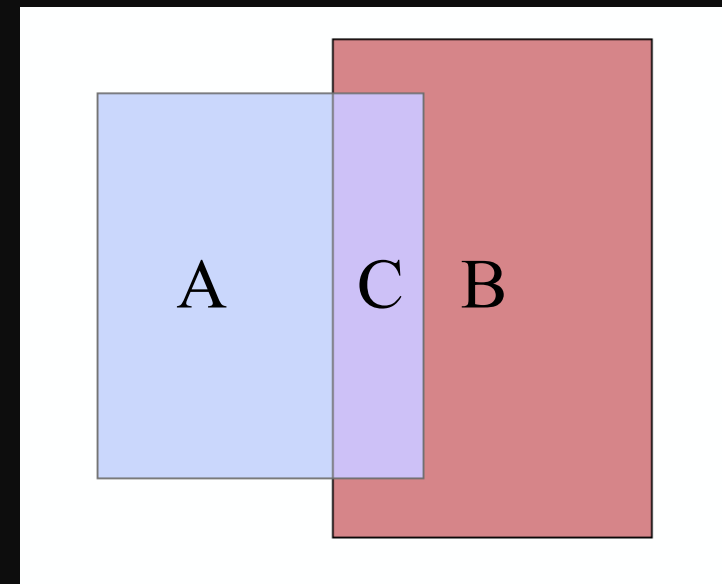
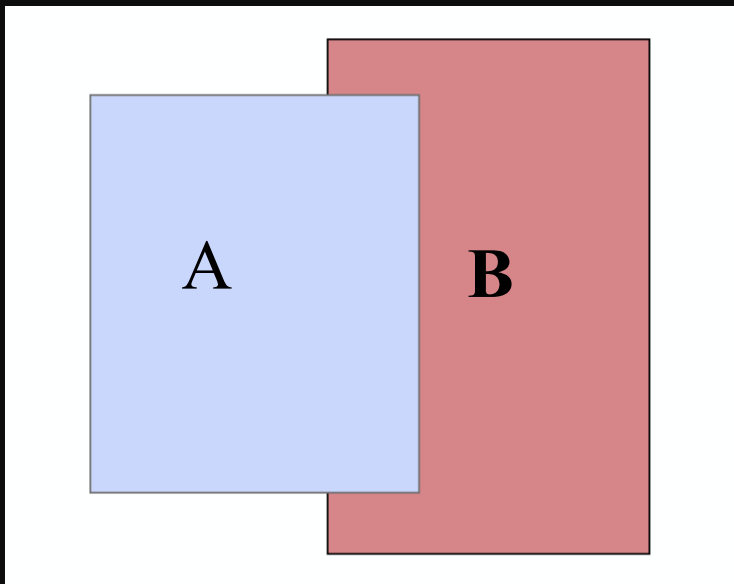


Intersect

Two polygons, A and B, Overlap.
Find the Intersection of A and B.

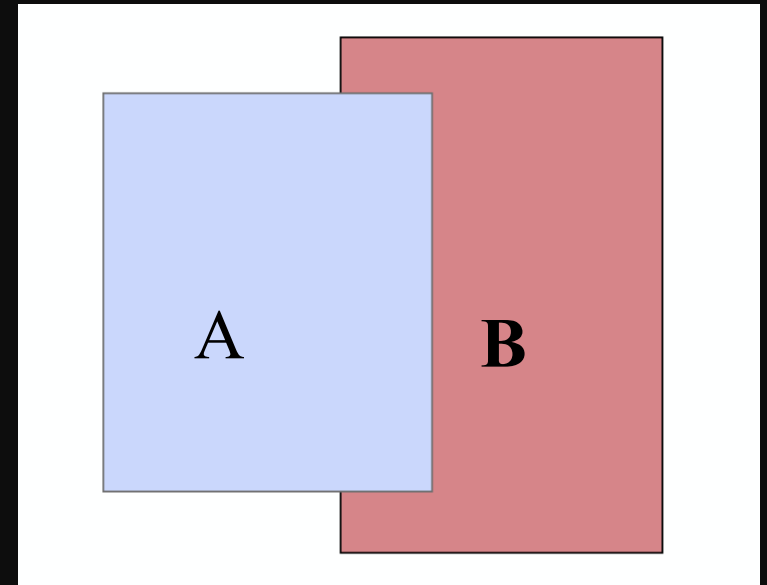
Intersect operation creates a new polygon, C, which is the intersection, or overlap, of A and B.

Attributes of A and B **both do appear** in C.



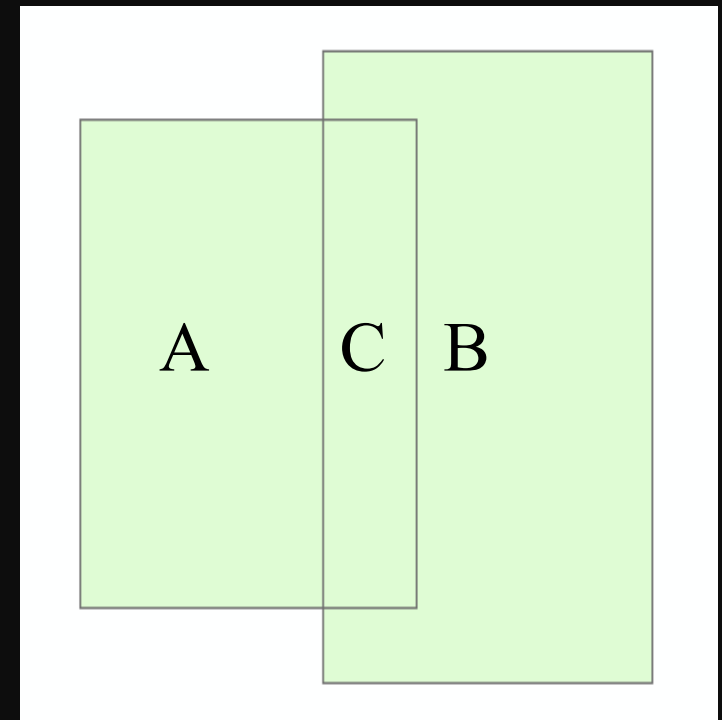
Union

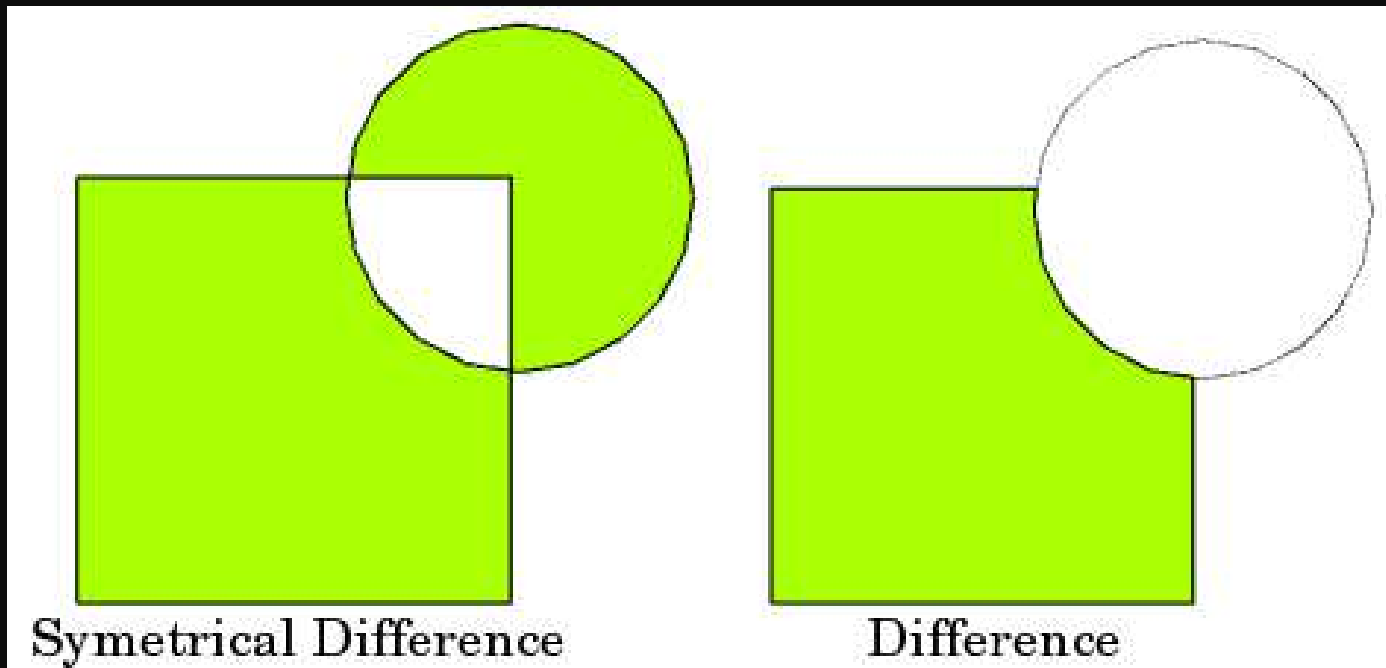
Two polygons, A and B, Overlap.
Find the union of A and B.



Union operation creates a new polygon, C, which is the intersect, or overlap, of A and B.

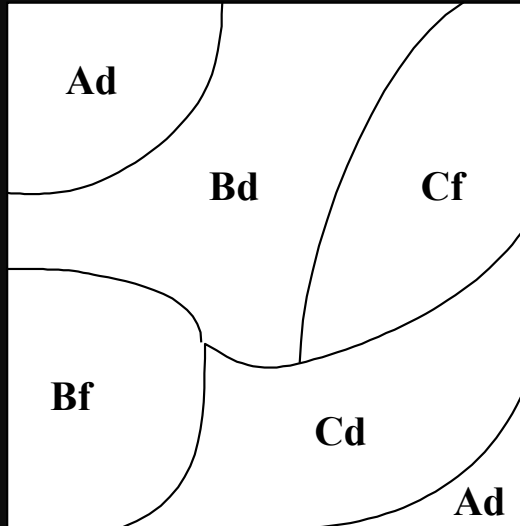
C contains the attributes of both A and B as they are part of the union.





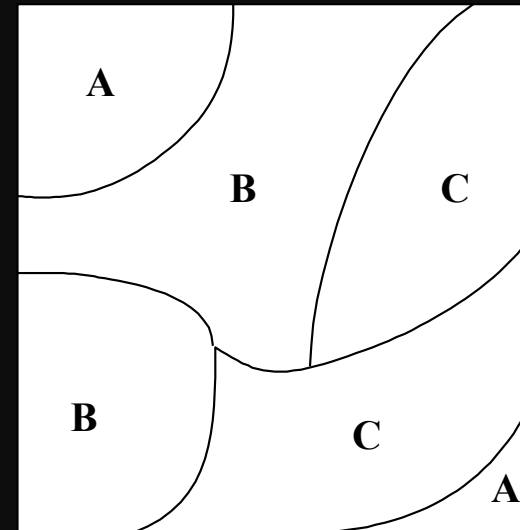
- **Symmetrical difference:** The output layer contains all areas of the input layers except those areas where the two layers overlap (intersect).
- **Difference:** The output layer contains all areas of the first input layer that do not overlap (intersect) with the second input layer.

Data Manipulation



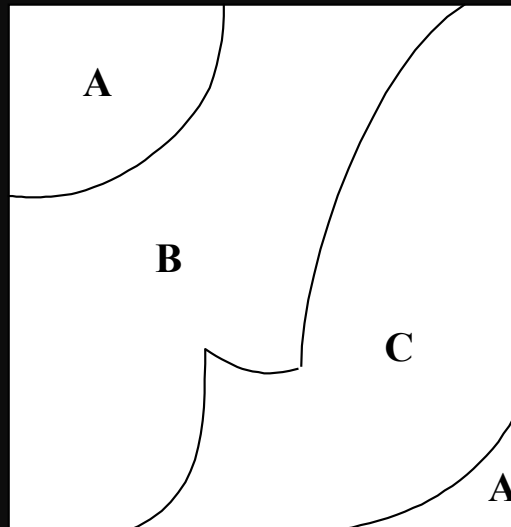
Soil types A, B and C with growth potentials d and f

Reclassify



Soil types A, B and C

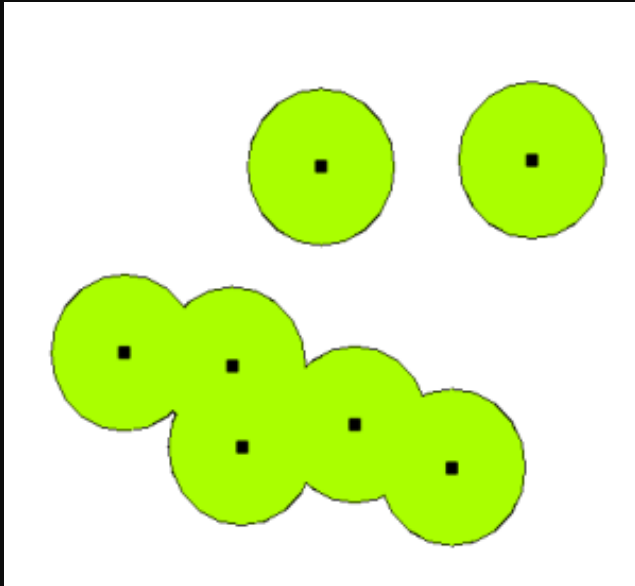
Dissolve & merge



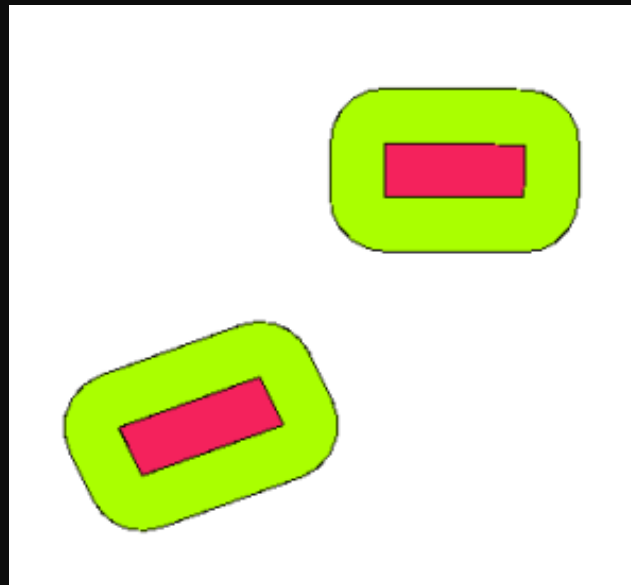
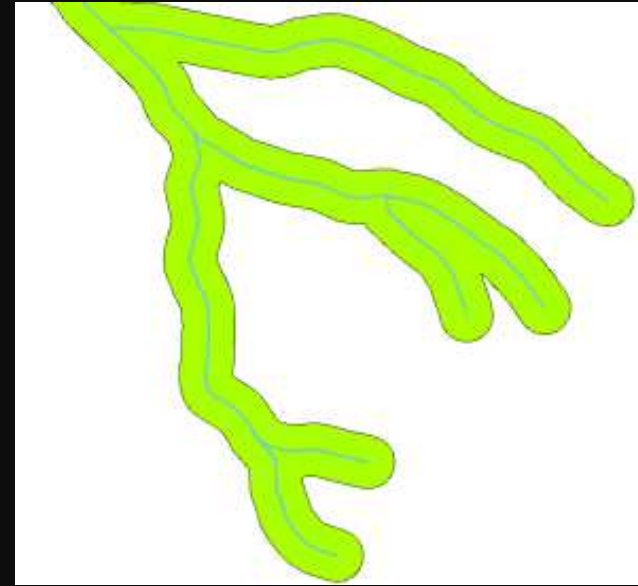
Soil types A, B and C

Buffering

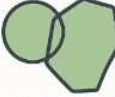
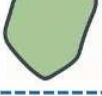
Buffering multi points



Buffering lines



Buffering an area
e.g. area within
100m of buildings.

Vector Geoprocessing Tool	What it Does (By Mashford Mahute)	Best Use Cases
 Buffer	Creates a zone of specified distance around input features (points, lines, or polygons).	Environmental impact zones, proximity analysis (e.g., find schools within 1 km of roads), noise pollution areas.
 Clip	Extracts input features that overlap with a clip feature (polygon).	Focus analysis on a specific area of interest, such as clipping roads to a municipal boundary.
 Intersect	Computes geometric intersection of input layers, retaining features common to all inputs.	Identifying overlapping land uses, zoning + flood zones, habitats intersecting with infrastructure.
 Union	Combines input polygon layers and retains all features and their attributes from both layers.	Land management analysis, overlaying administrative boundaries with ecological zones.
 Erase	Removes areas of the input layer that overlap with the erase layer.	Excluding protected areas from development zones, removing water bodies from land use layers.
 Merge	Combines multiple datasets of the same geometry type into a single output.	Integrating road networks from different districts into one layer, consolidating multiple forest patches.
 Dissolve	Aggregates features based on a shared attribute to remove internal boundaries.	Generalizing boundaries (e.g., districts into provinces), creating single features for analysis.
 Spatial Join	Joins attributes from one layer to another based on spatial relationships (e.g., proximity, containment).	Assigning population data from polygons to points, determining nearest facility.
 Select by Location	Selects features based on their spatial relationship with another layer.	Identifying buildings within flood zones, selecting farms within 500 m of a river.

Case Study: Identifying Flood-Exposed Villages, Schools and Roads

Problem statement

A district administration wants to identify villages, schools and road segments that are potentially exposed to flooding from a river, so that evacuation planning and emergency infrastructure can be prioritised

Data sets :River + villages + schools + roads + flood zone

Operations

Clip

Buffer zone

Intersect (village+ buffer zone)

Select schools

Intersect (road-flood zone)

Erase

Union (flood zone+ landuse polygons

Dissolve – continuous flood zone

Merge – road layers from different agencies